

Zhang, Xinmiao

professor

基本信息

姓名: Zhang, Xinmiao

所属院校: 俄亥俄州立大学-哥伦布

邮箱: zhang.8952@osu.edu

地区: United States

年龄范围: 40岁至50岁

企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

信息安全

网络空间安全

电子与计算机工程

集成电路设计与集成系统

通信工程

研究领域

密码学与密码分析

加密与数据保护

信息论与编码技术

数字集成电路设计

通信系统与网络 (电子计算机)

嵌入式系统与硬件设计

计算机体系结构

安全协议与安全认证

低功耗电路与能量管理

存储器设计与架构

个人简介

Xinmiao Zhang is a Professor in the Department of Electrical and Computer Engineering at The Ohio State University. She holds a Ph.D. in Electrical Engineering from the University of Minnesota, Twin Cities. Prof. Zhang's research focuses on VLSI architecture design, digital storage, communications,

hardware security, cryptography, and signal processing. She is a recipient of the NSF CAREER Award, IEEE Fellow, and has authored the book "VLSI Architectures for Modern Error-Correcting Codes." Prof. Zhang is currently the Associate Editor-in-Chief of IEEE Transactions on Circuits and Systems-I and has held various leadership roles in IEEE societies.

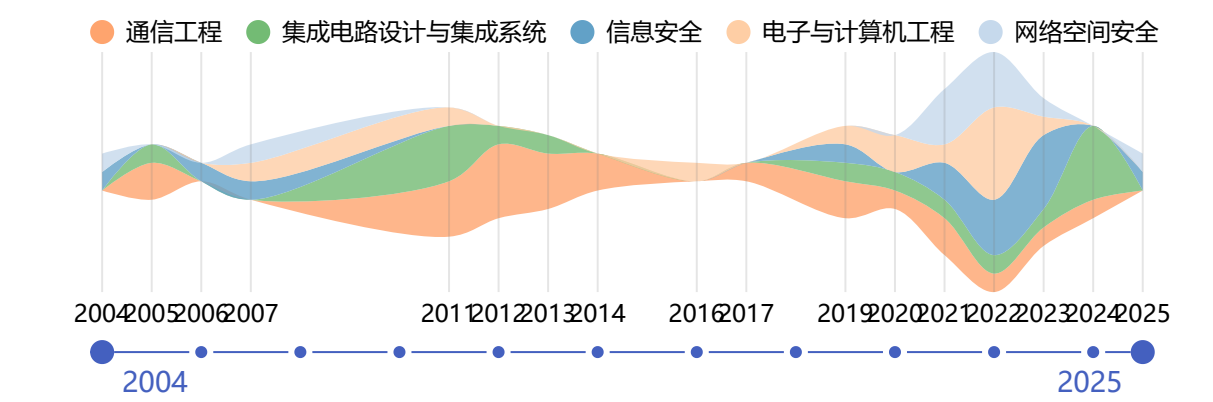
工作经历

Qualcomm - Visiting Position	- 2008
Case Western Reserve University - Assistant Professor	2005 - 2010
Case Western Reserve University - Associate Professor	2010 - 2013
University of Washington, Seattle - Visiting Position	2011 - 2013
IEEE Signal Processing Society - Chair	2013 - 2015
Western Digital/SanDisk - Senior Technologist	2013 - 2017
IEEE Communications Society - Vice-Chair	2017 - 2020
IEEE Circuits and Systems Society (CASS) - Board of Governors Member	2019 - 2021
IEEE Open Journal of Circuits and Systems (OJCAS) - Associate Editor	2019 -
Polytechnic University of Turin - Visiting Position	2020 -
IEEE Communications Society - Chair	2021 - 2022
IEEE Circuits and Systems Society (CASS) - Vice President - Technical Activities	2022 - 2023
IEEE Transactions on Circuits and Systems - I (TCAS - I) - Associate Editor - in - Chief	2024 - 2025
Ohio State University - Professor	- 至今
IEEE Transactions on Circuits and Systems - I (TCAS - I) - Associate Editor	2010 - 2019

教育经历

University of Minnesota, Twin Cities - Ph.D.	-
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近年发表的论文



Guest Editorial Special Issue on Emerging Hardware Security and Trust Technologies-AsianHOST 2023

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhang, Xinmiao; Gu, Chongyan; Ye, Mengmei; Liu, Qiang; Azarderakhsh, Reza; Liu, Weiqiang; Li, Yang

被引用次数: 02025

Guest Editorial Special Issue on the International Symposium on Integrated Circuits and Systems-ISICAS 2024

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhang, Xinmiao

被引用次数: 02024

Hardware Circuits and Systems Design for Post-Quantum Cryptography-A Tutorial Brief

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Xie, Jiafeng; Zhao, Wenfeng; Lee, Hanho; Roy, Debapriya Basu; Zhang, Xinmiao

被引用次数: 42024

Generalized Integrated Interleaved Codes for High-Density DRAMs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Tang, Yok Jye; Zhang, Xinmiao

被引用次数: 02024

Modified Extended Integrated Interleaved Codes

IEEE COMMUNICATIONS LETTERS

Tang, Yok Jye; Zhang, Xinmiao

被引用次数: 02024

Low-Complexity Parallel Min-Sum Medium-Density Parity-Check Decoder for McEliece Cryptosystem

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cai, Jiaxuan; Zhang, Xinmiao

被引用次数: 02023

High-Speed VLSI Architectures for Modular Polynomial Multiplication via Fast Filtering and Applications to Lattice-Based Cryptography

IEEE TRANSACTIONS ON COMPUTERS

Tan, Weihang; Wang, Antian; Zhang, Xinmiao; Lao, Yingjie; Parhi, Keshab K. K.

被引用次数: 6

2023

Joint Protection Scheme for Deep Neural Network Hardware Accelerators and Models

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhou, Jingbo; Zhang, Xinmiao

被引用次数: 3

2023

Algorithmic Obfuscation for LDPC Decoders

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhou, Jingbo; Zhang, Xinmiao

被引用次数: 0

2023

Efficient Check Node Processing for Min-Max NB-LDPC Decoding over Lower-Order Finite Fields

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Zhang, Xinmiao

被引用次数: 0

2022

Polynomial Multiplication Architecture with Integrated Modular Reduction for R-LWE Cryptosystems

JOURNAL OF SIGNAL PROCESSING SYSTEMS FOR SIGNAL IMAGE AND VIDEO TECHNOLOGY

Zhang, Xinmiao; Huai, Zheang; Parhi, Keshab K.

被引用次数: 2

2022

Efficient Hardware Implementation Architectures for Long Integer Modular Multiplication over General Solinas Prime

JOURNAL OF SIGNAL PROCESSING SYSTEMS FOR SIGNAL IMAGE AND VIDEO TECHNOLOGY

Huai, Zheang; Zhou, Jingbo; Zhang, Xinmiao

被引用次数: 2

2022

Low-Complexity Resource-Shareable Parallel Generalized Integrated Interleaved Encoder

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Tang, Yok Jye; Zhang, Xinmiao

被引用次数: 6

2022

Efficient Reconfigurable Vandermonde Matrix Inverter for Erasure-Correcting Generalized Integrated Interleaved Decoding

2022 IEEE WORKSHOP ON SIGNAL PROCESSING SYSTEMS (SIPS)

Tang, Yok Jye; Zhang, Xinmiao

被引用次数: 0

2022

Low-Complexity AES Architectures Resilient to Power Analysis Attacks

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Zhou, Jingbo; Elgendy, Elsayed; Tawfik, Eslam Yahya; Zhang, Xinmiao

被引用次数: 1

2022

Fast En/Decoding of Reed-Solomon Codes for Failure Recovery

IEEE TRANSACTIONS ON COMPUTERS

Tang, Yok Jye; Zhang, Xinmiao

被引用次数: 10

2022

Efficient Nested Key Equation Solver for Short Generalized Integrated Interleaved BCH Codes

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Xie, Zhenshan; Zhang, Xinmiao

被引用次数: 1

2022

A Survey on High-Throughput Non-Binary LDPC Decoders: ASIC, FPGA, and GPU Architectures

IEEE COMMUNICATIONS SURVEYS AND TUTORIALS

Ferraz, Oscar; Subramaniyan, Srinivasan; Chinthala, Ramesh; Andrade, Joao; Cavallaro, Joseph R.; Nandy, Soumitra K.; Silva, Vitor; Zhang, Xinmiao; Purnaprajna, Madhura; Falcao, Gabriel

被引用次数: 24

2022

Fast Nested Key Equation Solvers for Generalized Integrated Interleaved Decoder

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xie, Zhenshan; Zhang, Xinmiao

被引用次数: 9

2021

Pipelined High-throughput NTT Architecture for Lattice-Based Cryptography

PROCEEDINGS OF THE 2021 ASIAN HARDWARE ORIENTED SECURITY AND TRUST SYMPOSIUM (ASIANHOST)

Tan, Weihang; Wang, Antian; Lao, Yingjie; Zhang, Xinmiao; Parhi, Keshab K.

被引用次数: 6

2021

A Low-Complexity Flexible Logic-Locking Scheme Resisting Removal Attacks

2021 IEEE INTERNATIONAL MIDWEST SYMPOSIUM ON CIRCUITS AND SYSTEMS (MWSCAS)

Zhou, Jingbo; Zhang, Xinmiao

被引用次数: 0

2021

Efficient Architecture for Long Integer Modular Multiplication over Solinas Prime

2021 IEEE WORKSHOP ON SIGNAL PROCESSING SYSTEMS (SIPS 2021)

Huai, Zheang; Parhi, Keshab K.; Zhang, Xinmiao

被引用次数: 1

2021

An Efficient Parallel Architecture for Resource-Shareable Reed-Solomon Encoder

2021 IEEE WORKSHOP ON SIGNAL PROCESSING SYSTEMS (SIPS 2021)

Tang, Yok Jye; Zhang, Xinmiao

被引用次数: 2	2021
Miscorrection Mitigation for Generalized Integrated Interleaved BCH Codes	
IEEE COMMUNICATIONS LETTERS	
Xie, Zhenshan; Zhang, Xinmiao	
被引用次数: 8	2021
Guest Editorial: Special Section on the Design and Implementation of Signal Processing Systems	
JOURNAL OF SIGNAL PROCESSING SYSTEMS FOR SIGNAL IMAGE AND VIDEO TECHNOLOGY	
Zhang, Xinmiao; Wang, Zhongfeng; Studer, Christoph	
被引用次数: 0	2021
Generalized SAT-Attack-Resistant Logic Locking	
IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY	
Zhou, Jingbo; Zhang, Xinmiao	
被引用次数: 15	2021
Efficient VLSI Architectures for Coupled-Layered Regenerating Codes	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Zhang, Xinmiao; Xie, Zhenshan	
被引用次数: 1	2020
Scaled Nested Key Equation Solver for Generalized Integrated Interleaved Decoder	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Xie, Zhenshan; Zhang, Xinmiao	
被引用次数: 7	2020
Reduced-Complexity Key Equation Solvers for Generalized Integrated Interleaved BCH Decoders	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Xie, Zhenshan; Zhang, Xinmiao	
被引用次数: 9	2020
Efficient Architectures for Generalized Integrated Interleaved Decoder	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Zhang, Xinmiao; Xie, Zhenshan	
被引用次数: 17	2019
Relaxing the Constraints on Locally Recoverable Erasure Codes by Finite Field Element Variation	
IEEE COMMUNICATIONS LETTERS	
Zhang, Xinmiao; Xie, Zhenshan	
被引用次数: 6	2019
On the Construction of Composite Finite Fields for Hardware Obfuscation	
IEEE TRANSACTIONS ON COMPUTERS	

Zhang, Xinmiao; Lao, Yingjie

被引用次数: 6

2019

A Low-Power Parallel Architecture for Linear Feedback Shift Registers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Xinmiao

被引用次数: 21

2019

Generalized Three-Layer Integrated Interleaved Codes

IEEE COMMUNICATIONS LETTERS

Zhang, Xinmiao

被引用次数: 20

2018

Modified Generalized Integrated Interleaved Codes for Local Erasure Recovery

IEEE COMMUNICATIONS LETTERS

Zhang, Xinmiao

被引用次数: 21

2017

Low-Complexity Transformed Encoder Architectures for Quasi-Cyclic Nonbinary LDPC Codes Over Subfields

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Xinmiao; Tai, Ying

被引用次数: 4

2017

A Flexible and Low-Complexity Local Erasure Recovery Scheme

IEEE COMMUNICATIONS LETTERS

Zhang, Xinmiao; Sprouse, Steven; Ilani, Ishai

被引用次数: 11

2016

Finite Alphabet Iterative Decoders for LDPC Codes: Optimization, Architecture and Analysis

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cai, Fang; Zhang, Xinmiao; Declercq, David; Planjery, Shiva Kumar; Vasic, Bane

被引用次数: 15

2014

Efficient Check Node Processing Architectures for Non-binary LDPC Decoding Using Power Representation

JOURNAL OF SIGNAL PROCESSING SYSTEMS FOR SIGNAL IMAGE AND VIDEO TECHNOLOGY

Cai, Fang; Zhang, Xinmiao

被引用次数: 1

2014

Generalized Backward Interpolation for Algebraic Soft-Decision Decoding of Reed-Solomon Codes

IEEE TRANSACTIONS ON COMMUNICATIONS

Zhang, Xinmiao; Zheng, Yu

被引用次数: 5

2013

An Efficient Interpolation-Based Chase BCH Decoder

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Xinmiao

被引用次数: 9

2013

Relaxed Min-Max Decoder Architectures for Nonbinary Low-Density Parity-Check Codes

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Cai, Fang; Zhang, Xinmiao

被引用次数: 29

2013

A Low-Complexity Three-Error-Correcting BCH Decoder for Optical Transport Network

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Xinmiao; Wang, Zhongfeng

被引用次数: 27

2012

Efficient Generalized Minimum-distance Decoders of Reed-Solomon Codes

JOURNAL OF SIGNAL PROCESSING SYSTEMS FOR SIGNAL IMAGE AND VIDEO TECHNOLOGY

Zhu, Jiangli; Zhang, Xinmiao

被引用次数: 0

2012

Novel Interpolation and Polynomial Selection for Low-Complexity Chase Soft-Decision Reed-Solomon Decoding

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Xinmiao; Wu, Yingquan; Zhu, Jiangli; Zheng, Yu

被引用次数: 1

2012

Low-Complexity Reliability-Based Message-Passing Decoder Architectures for Non-Binary LDPC Codes

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Xinmiao; Cai, Fang; Lin, Shu

被引用次数: 43

2012

Reduced-Complexity Decoder Architecture for Non-Binary LDPC Codes

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Xinmiao; Cai, Fang

被引用次数: 66

2011

Reduced-complexity column-layered decoding and implementation for LDPC codes

IET COMMUNICATIONS

Cui, Z.; Wang, Z.; Zhang, X.

被引用次数: 37

2011

Reliability-Driven ECC Allocation for Multiple Bit Error Resilience in Processor Cache

IEEE TRANSACTIONS ON COMPUTERS

Paul, Somnath; Cai, Fang; Zhang, Xinmiao; Bhunia, Swarup

被引用次数: 41

2011

Efficient Partial-Parallel Decoder Architecture for Quasi-Cyclic Nonbinary LDPC Codes

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhang, Xinmiao; Cai, Fang

被引用次数: 45

2011

MONET special issue on next generation hardware architectures for secure mobile computing

MOBILE NETWORKS & APPLICATIONS

Sklavos, Nicolas; McLoone, Maire; Zhang, Xinmiao

被引用次数: 0

2007

On the optimum constructions of composite field for the AES algorithm

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Xinmiao; Parhi, Keshab K.

被引用次数: 76

2006

Fast factorization architecture in soft-decision Reed-Solomon decoding

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, XM; Parhi, KK

被引用次数: 25

2005

High-speed architectures for parallel long BCH encoders

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, XM; Parhi, KK

被引用次数: 45

2005

High-speed VLSI architectures for the AES algorithm

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, XM; Parhi, KK

被引用次数: 275

2004

On the Construction of Composite Finite Fields for Hardware Obfuscation

TC

Xinmiao Zhang; Yingjie Lao

被引用次数: 0

Generalized SAT-Attack-Resistant Logic Locking

TIFS

Jingbo Zhou; Xinmiao Zhang

被引用次数: 0

Algorithmic Obfuscation for LDPC Decoders

TCAD

Jingbo Zhou; Xinmiao Zhang

被引用次数: 0

Fast En/Decoding of Reed-Solomon Codes for Failure Recovery

TC

Yok Jye Tang; Xinmiao Zhang

被引用次数: 0

学术合作



Zhenshan XIE	合作次数: 4
俄亥俄州立大学-哥伦布	
Keshab K. Parhi	合作次数: 3
明尼苏达大学-双城	
刘伟强	合作次数: 1
南京航空航天大学	
王中风	合作次数: 1
南京大学	
Joseph R. Cavallaro	合作次数: 1
莱斯大学	

还有 3 个合作者